

Sno	Author Names	Title of the Paper	Year of Published	Dataset Used
1	Serdar Gündoğdu	Efficient prediction of early-stage diabetes using XGBoost classifier with random forest feature selection technique	2023	Sylhet Diabetes Hospital dataset, Bangladesh (520 records)
2	Ayşe Doğru; Selim Buyrukoğlu; Murat Arı	A hybrid super ensemble learning model for the early-stage prediction of diabetes risk	2023	Early-stage diabetes risk dataset; PIMA; Diabetes 130-US hospitals
3	Liyang Wang; Xiaoya Wang; Angxuan Chen; Xian Jin; Huilian Che	Prediction of Type 2 Diabetes Risk and Its Effect Evaluation Based on the XGBoost Model	2020	Questionnaire data from 380 middle-aged/elderly participants in Xicheng District, Beijing
4	Nurjahan Nipa; Mahmudul Hasan Riyad; Shahriare Satu; Walliullah; Koushik Chandra Howlader; Mohammad Ali Moni	Clinically adaptable machine learning model to identify early appreciable features of diabetes	2024	Sylhet Diabetes Hospital dataset (520); additional Bangladesh questionnaire dataset (558); combined data
5	Aishwariya Dutta; Md. Kamrul Hasan; Mohiuddin Ahmad; Md. Abdul Awal; Md. Akhtarul Islam; Mehedi Masud; Hossam Meshref	Early Prediction of Diabetes Using an Ensemble of Machine Learning Models	2022	Newly labeled South Asian diabetes dataset

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6	Karlo Abnoosian; Rahman Farnoosh; Mohammad Hassan Behzadi	Prediction of diabetes disease using an ensemble of machine learning multi-classifier models	2023	Iraqi Patient Dataset of Diabetes
7	Xin Feng; Yihuai Cai; Ruihao Xin	Optimizing diabetes classification with a machine learning-based framework	2023	Diabetes classification dataset(s) used in the study
8	Victor Chang; Jozeene Bailey; Qianwen Ariel Xu; Zhili Sun	Pima Indians diabetes mellitus classification based on machine learning (ML) algorithms	2022	Pima Indians Diabetes Dataset
9	Michael Onyema Edeh; Osamah Ibrahim Khalaf; Carlos Andrés Tavera; Sofiane Tayeb; Saad Ghouali; G. M. Abdulsahib; Nnamdi E. Richard-Nnabu; Abdelouahed Louni	A Classification Algorithm-Based Hybrid Diabetes Prediction Model	2022	Frankfurt Hospital diabetes dataset and Pima Indians Diabetes Database
10	Umair Muneer Butt; Sukumar Letchmunan; Mubashir Ali; Fadratul Hafinaz Hassan; Anees Baqir; Hafiz Husnain Raza Sherazi	Machine Learning Based Diabetes Classification and Prediction for Healthcare Applications	2021	PIMA Indian Diabetes dataset

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11	Huma Naz; Sachin Ahuja	Deep learning approach for diabetes prediction using PIMA Indian dataset	2020	PIMA Indian Diabetes dataset
12	Roshi Saxena; Sanjay Kumar Sharma; Manali Gupta; G. C. Sampada	A Novel Approach for Feature Selection and Classification of Diabetes Mellitus: Machine Learning Methods	2022	PIMA Indians Diabetes dataset
13	Qazi Waqas Khan; Khalid Iqbal; Rashid Ahmad; Atif Rizwan; Anam Nawaz Khan; DoHyeun Kim	An intelligent diabetes classification and perception framework based on ensemble and deep learning method	2024	PIMA dataset and early-risk diabetes dataset
14	Shimoo Firdous; Gowher A. Wagai; Kalpana Sharma	A survey on diabetes risk prediction using machine learning approaches	2022	Multiple datasets reported across PubMed, IEEE Xplore, INSPEC and other sources
15	Farida Mohsen; Hamada R. H. Al-Absi; Noha A. Yousri; Nady El Hajj; Zubair Shah	A scoping review of artificial intelligence-based methods for diabetes risk prediction	2023	Multiple datasets/modalities across 40 included studies

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16	Pir Bakhsh Khokhar; Carmine Gravino; Fabio Palomba	Advances in artificial intelligence for diabetes prediction: insights from a systematic literature review	2025	Multiple diabetes datasets including PIDD, NHANES and other healthcare datasets
17	Saleha Masood; Mousa Ahmad Al-Bashrawi; Muhammad Attique Khan; Yogesh K. Dwivedi et al.	From data to diagnosis: a systematic review on AI-driven approaches to diabetes prediction	2026	Multiple datasets and modalities reported in 155 studies
18	Gufran Ahmad Ansari; Salliah Shafi; Mohd Dilshad Ansari; Azhar Shadab	Advanced supervised machine learning methods for precise diabetes mellitus prediction using feature selection	2025	Pima Indian Diabetes dataset
19	Honghao Li; Dongqing Su; Xinpeng Zhang; Yuanyuan He; Xu Luo; Yuqiang Xiong; Min Zou; Huiyan Wei; Shaoran Wen; Qilemuge Xi; Yongchun Zuo; Lei Yang	Machine learning-based prediction of diabetic patients using blood routine data	2024	Hospital/health-center blood routine data: 1,000 diabetes patients and 1,000 healthy individuals
20	Abeer El-Sayyid El-Bashbishy; Hazem M. El-Bakry	Pediatric diabetes prediction using machine learning	2026	Integrated pediatric/diabetes datasets including PIMA and other sources

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21	Ilkim Ecem Emre	Predicting Type 2 Diabetes Using Random Forest and XGBoost Algorithms: A Comparative Machine Learning Approach	2026	PIMA Indians Diabetes dataset
22	Angga Ramadhan; Artha Maulana Rahman; Rafa Muhammad Rayhan; Dani Rofianto; Khusnatul Amaliah	Comparative Analysis of Gradient Boosting, Random Forest, and XGBoost Algorithms in Diabetes Mellitus Prediction	2026	Pima Indians Diabetes dataset
23	Aishwariya Dutta; Md. Kamrul Hasan; Mohiuddin Ahmad; Md. Abdul Awal; Md. Akhtarul Islam; Mehedi Masud; Hossam Meshref	Early Prediction of Diabetes Using an Ensemble of Machine Learning Models	2022	South Asian diabetes dataset
24	Breiman, Leo	Random Forests	2001	Benchmark datasets; not diabetes-specific
25	Tianqi Chen; Carlos Guestrin	XGBoost: A Scalable Tree Boosting System	2016	Multiple benchmark datasets; not diabetes-specific

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26	J. W. Smith; J. E. Everhart; W. C. Dickson; W. C. Knowler; R. S. Johannes	Using the ADAP learning algorithm to forecast the onset of diabetes mellitus	1988	Pima Indians Diabetes dataset

Methodology Used	Limitation of the Paper
MLR + Random Forest feature selection; XGBoost classification; preprocessing; evaluation with accuracy, precision, recall, F1-score and AUC.	Uses a relatively small, questionnaire-based dataset from one setting; external validation/generalization to other populations is limited.
Chi-square feature selection; logistic regression, decision tree, random forest and gradient boosting base learners with SVM meta-learner; GridSearch.	The approach is more complex than a two-model comparison and still requires broader external validation before clinical deployment.
XGBoost compared with SVM, Random Forest and KNN; 10-fold cross-validation.	Small, geographically localized sample may limit generalization to other populations.
35 classification algorithms including Random Forest and XGBoost; preprocessing; feature interpretation using SHAP; accuracy, precision, recall, F1-score and AUC.	Data were collected from specific Bangladesh populations; stability across wider populations needs further validation.
Missing-value imputation, outlier handling, feature selection, K-fold cross-validation; weighted ensemble of NB, RF, DT, XGBoost and LightGBM; GridSearch.	The newly collected dataset is relatively limited in size and population coverage; wider validation is needed.

Methodology Used	Limitation of the Paper
Data cleaning, imputation, normalization/standardization, feature selection, k-fold CV; k-NN, SVM, DT, RF, AdaBoost and GNB; weighted ensemble and hyperparameter optimization.	The dataset is imbalanced and population-specific; further validation on independent populations is needed.
Machine-learning framework with optimized GAN; preprocessing and synthetic-data generation to improve classification.	Synthetic data generation can introduce distribution differences from real clinical data; generalization requires additional external datasets.
Naive Bayes, Random Forest and J48 decision tree; comparison using accuracy, precision, sensitivity and specificity; interpretable ML focus.	Pima dataset is relatively small and population-specific; results may not generalize to other demographic groups.
Random Forest, SVM, Naive Bayes, Decision Tree and K-means; training/test split and comparative classification.	Performance differs by dataset, indicating sensitivity to dataset characteristics; broader external validation is needed.
Random Forest, MLP and Logistic Regression for classification; LSTM, moving averages and linear regression for prediction.	Uses a benchmark dataset and a hypothetical IoT monitoring concept; real-world clinical deployment was not demonstrated.

Methodology Used	Limitation of the Paper
ANN, Naive Bayes, Decision Tree and deep learning models; comparative evaluation.	Small benchmark dataset limits generalization; deep learning performance may depend strongly on preprocessing and data size.
Preprocessing, outlier removal, mean imputation, feature selection and hyperparameter optimization; MLP, DT, KNN and RF.	The study relies on the PIMA dataset, so external population validation is limited.
Chi-square, mutual information and sequential feature selection; ANN, RF, gradient boosting, TabNet and SVM.	Feature-selection and model combinations increase complexity; independent clinical validation is still required.
Literature survey of supervised and unsupervised ML approaches including SVM, KNN and Random Forest.	It is a survey rather than a new predictive model, so it does not provide a single validated implementation for deployment.
PRISMA-based scoping review covering classical ML, deep learning, EHR, multi-omics and imaging; emphasis on validation and interpretability.	Most reviewed studies used internal validation; only a small number used external validation, limiting evidence for generalizability.

Methodology Used	Limitation of the Paper
Systematic review of 53 studies covering SVM, XGBoost, logistic regression, CNN and explainable AI methods.	The review identifies recurring issues including class imbalance, interpretability and the need for clinically validated explainable models.
Systematic review of AI/ML/DL approaches, including Random Forest and XGBoost, with emphasis on performance, generalization and clinical interpretability.	Heterogeneity of datasets and evaluation methods makes direct comparison across studies difficult.
Feature selection and 10-fold cross-validation; SVM, Naive Bayes, KNN and Random Forest; accuracy, precision, recall and F1-score.	Evaluation is based on the Pima dataset; broader demographic validation is needed.
XGBoost, Random Forest, SVM and Elastic Net; external validation and SHAP-based interpretation.	Blood-routine features and a specific clinical population may limit transferability to settings with different patient characteristics.
Supervised ML models including ANN, Logistic Regression, Naive Bayes, DT, AdaBoost, RF, Gradient Boosting, SVM and KNN.	Heterogeneous source datasets can introduce differences in feature definitions and distributions; further prospective validation is needed.

Methodology Used	Limitation of the Paper
Preprocessing and repeated 5-fold cross-validation; Random Forest and XGBoost compared using accuracy, precision, sensitivity, specificity, F1-score and AUC.	Uses the PIMA benchmark dataset; performance may not directly generalize to other populations or clinical settings.
Preprocessing with median imputation and Min-Max scaling; comparison of Gradient Boosting, Random Forest and XGBoost.	Dataset is a benchmark dataset with known population limitations; external validation is required for broader use.
Weighted ensemble using NB, RF, DT, XGBoost and LightGBM; GridSearch, feature selection, imputation and K-fold CV.	Limited labeled clinical data and missing/outlier values remain challenges; wider population validation is needed.
Introduced Random Forest ensemble learning using multiple randomized decision trees and aggregation.	Foundational method is not diabetes-specific and does not address healthcare-specific data quality or clinical validation.
Introduced scalable gradient-boosted decision trees with regularization and efficient training.	Methodology paper is not diabetes-specific; clinical interpretability and healthcare validation are outside its scope.

Methodology Used	Limitation of the Paper
ADAP learning algorithm for diabetes onset prediction using clinical attributes.	Older and relatively small population-specific dataset; methodology predates modern ensemble ML and may not reflect current clinical data.